

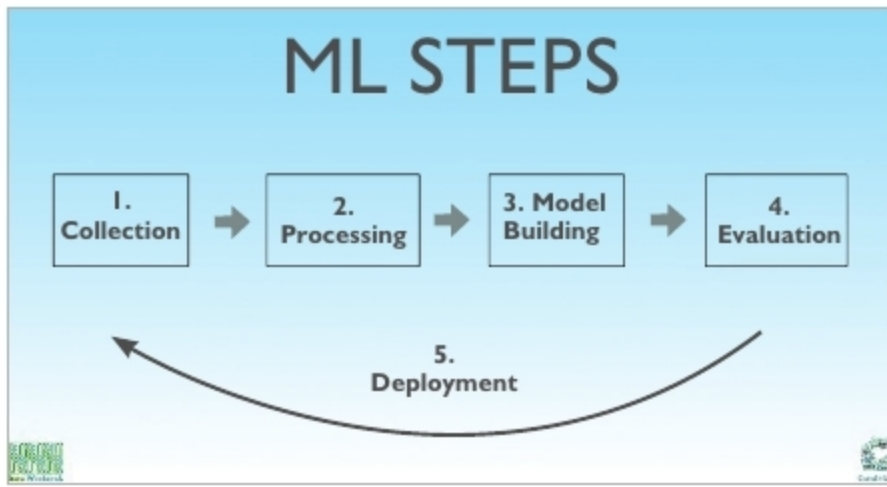


Figure 2: Different steps involved in developing a machine learning project (Image source: *googleimages.com*)

4. Python has got considerable support from the community, so developers can easily find a large number of tutorials and valuable tips during the development process. This makes it easier to use any new technology from scratch.
5. Python is like the new FORTRAN of the scientific world. It's very popular in the non-computer scientist's world, equipping users with an enormously large toolbox for different kinds of applied programming problems.
6. Python makes it very easy to quickly implement or experiment with any new ideas and prototypes. Hence, different scientific and research communities love to use it. That is why it is being widely used in ML and data science.

Starting a Python machine learning project

The following steps act as a good guide for how to begin work on a machine learning project.

1. **Define the problem:** The problem or the scenario for which we want to develop an ML solution must be defined and understood. At times, there is a deviation from the pre-defined problem set while the ML solution is being developed. Hence, this is a significant step to start with.
2. **Prepare the data:** The data sets for a ML solution that have to be processed and analysed to give a concluding result must be prepared. These available data sets are the inputs used by algorithms and several analysis operations are performed on them before coming to a final result. These data sets have a great impact on the final result.
3. **Evaluate algorithms:** Different algorithms must be analysed before choosing and deciding to go ahead with one. The algorithm chosen decides the performance of the ML project. The problem set, the behaviour and the volume of the data set, as well as what is expected of the ML project must be considered before choosing an algorithm. There is every possibility that more than two algorithms may perform operations on the data set and give the desired result but the performance factor associated with each algorithm must be evaluated. This plays quite a significant role, especially when it comes to a large ML project.

4. **Improve the results:** As any ML project has the basic quality of continuously improving itself in terms of the technique used by it or the algorithm implemented, it always obtains better results, every time. Also, machines learn from the diverse set of data which they analyse before giving the result. Hence, even the quality of data sets helps an ML solution produce improved results.
5. **Present the results:** Once the machine fetches the desired result, it is very important that it is presented in an effective manner so that it can be used further for different applications. The presentation of the desired result is one of the very basic qualities of any ML project. The results can be presented with the help of graphical plots, numerically in the form of some table or even just as the concluding result. But, ultimately, the project should present the result in the desired manner.

Now let us consider some other important factors to ensure an effective and efficient ML project.

Model building, training and evaluation: Developing a machine learning model is a very responsible task as it is almost similar to building a product. The starting point is always ideation, wherein the problem for which the solution is required is considered in its various aspects with some potential approaches. Once a clear direction is established, the solution is prototyped and further tested to check if it meets the needs. A continuous loop between the process of ideation, prototyping and validation takes place until the ML solution is mature enough to be brought into the market, when it is productised for a broader launch.

Data pipeline: A machine learning algorithm generally takes a cleaned data set and learns some patterns in it so that it can decide and make some predictions on the new data set. However, when ML is used for real-life applications, the raw data obtained from the real world is not ready to be directly fed into the ML algorithm. This raw data needs to be preprocessed to generate input data for the ML algorithm. Hence, this entire process of converting the raw data to usable data by the ML algorithm, training an ML algorithm, and finally using this output to perform different actions in the real world is called a pipeline.

Data wrangling: To avoid the ‘garbage-in-garbage-out’ (GIGO) condition, mapping and the transformation of raw data into ML-ready data must be done properly. Data wrangling is one of the key processes that enables an ML project to make a big impact by avoiding GIGO situations. Filtering, cleansing, joining and stacking are the necessary steps to get data ready for ML. Pyspark is one of the widely used engines for data wrangling. This can also be seamlessly integrated with other Big Data engines. Python has got Dask as its indigenous Big Data engine.

Feature extraction and engineering: In an ML project, feature extraction has got its own role to play. During different phases like image processing and pattern recognition in a machine learning project, feature extraction starts from a very initial set of measured data and builds derived values (also